Software Test Plan (STP)
Online Internship & Placement Preparation Portal (IPP)

Project: Online Internship & Placement Preparation Portal (IPP)

Version: 1.0

Authors: PES2UG24CS186, PES2UG24CS190, PES2UG24CS208, PES2UG24CS214 (QA Team)

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1. Introduction

Purpose: This document defines the test plan for the Online Internship & Placement Preparation Portal (IPP) v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing the system described in the Software Requirements Specification (SRS).

Scope: Testing covers IPP features such as role-based authentication, internship listing management and search, application management, the timed aptitude/coding test engine, progress tracking and placement analytics dashboards, and the admin/placement-officer module. Payroll/HR onboarding after selection, payment processing, and integration with third-party applicant tracking systems (ATS) are excluded, as they are out of scope of the SRS.

References: IPP SRS v1.0, MERN stack design specifications, institutional data-privacy norms, WCAG 2.1 AA guidelines.

Definitions: IPP (Internship & Placement Preparation Portal), MERN (MongoDB, Express.js, React.js, Node.js), JWT (JSON Web Token), RBAC (Role-Based Access Control), OTP (One-Time Password), API (Application Programming Interface), CSV (Comma-Separated Values), TLS (Transport Layer Security), WCAG (Web Content Accessibility Guidelines), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix).

2. Test Items

- Authentication & User Management module
- Internship Listings & Search module
- Application Management module
- Aptitude Test module (Node.js API + Python scoring microservice)
- Progress Tracking & Analytics Dashboard module
- Admin & Placement Officer module

3. Features to be Tested

Features mapped to SRS requirement IDs:

- IPP-F-001 / IPP-F-002 / IPP-F-003: Role-based registration, JWT authentication, password hashing
- IPP-F-004 / IPP-F-005: Account lockout and password reset
- IPP-F-006 / IPP-F-007 / IPP-F-009: Listing creation, search & filter, deadline auto-hide
- IPP-F-010 / IPP-F-011 / IPP-F-012: Resume upload & apply, application status update, notifications
- IPP-F-014 / IPP-F-015 / IPP-F-016: Timed aptitude test, auto-evaluation, reattempt restriction
- IPP-F-018 / IPP-F-019 / IPP-F-021: Student dashboard, placement analytics, CSV/PDF export
- IPP-F-022 / IPP-F-023: Recruiter approval workflow, RBAC enforcement
- IPP-NF-001: API/page response time ≤ 8s for 90% of requests
- IPP-NF-002: 90% monthly uptime
- IPP-SR-002 / IPP-SR-006: JWT lifetime & revocation, RBAC at API layer

4. Features Not to be Tested

- Payroll/HR onboarding after candidate selection — outside the scope of the current system.
- Payment processing — no payment-related functionality is included in the current release.
- Third-party Applicant Tracking System (ATS) integrations — ATS integration is planned for a future release.
- Internal implementation of third-party email/SMS providers — only the portal's integration with these services will be tested.
- Internal implementation of cloud/local file storage — resume upload, validation, and retrieval will be tested, but the storage provider's internal operations will not.

5. Test Approach / Strategy

Levels

- Unit tests (module-level: auth, listings, applications, test engine, analytics)
- Integration tests (Node.js API ↔ MongoDB, Node.js API ↔ Python analytics/scoring microservice)
- System tests (end-to-end IPP functionality across Student, Recruiter, Admin roles)
- Acceptance tests (UAT with representative students, recruiters, and placement officers)

Types

- Functional testing (core features across all modules)
- Regression testing (after each build/release)
- Performance testing (response time and load, per IPP-NF-001 and IPP-NF-005)
- Usability and accessibility testing (WCAG 2.1 AA conformance, per IPP-NF-004)
- Security testing (see Section 5.1)

Entry Criteria: Stable build delivered, test data (seeded student/recruiter/admin accounts) available, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all High-priority functional requirements verified, and the RTM (Section 13) shows all mapped test cases passed.

5.1 Security Validation

- Validate password hashing (bcrypt/argon2) and confirm no plaintext storage (IPP-F-003)
- TLS 1.2+ verification for all client-server and server-to-microservice traffic (IPP-SR-001)
- JWT lifetime and refresh-token revocation checks (IPP-SR-002)
- Input validation/sanitization checks for NoSQL injection and XSS (IPP-SR-003)
- Rate-limiting checks on authentication and file-upload endpoints (IPP-SR-004)
- Resume upload validation: file type, size limit, and malicious-content scanning (IPP-SR-005)
- RBAC enforcement checks across all API endpoints (IPP-SR-006)

6. Test Environment

Hardware: Standard client devices (desktop/laptop/mobile) with modern browsers and internet access; cloud/on-premise servers hosting the Node.js API, MongoDB instance, and Python analytics service.

Software: React.js front end, Node.js/Express REST API, MongoDB (via Mongoose), Python (Flask) analytics/scoring microservice, third-party email/SMS API sandbox.

Tools: Selenium/Cypress (UI automation), Postman (API testing), JMeter/k6 (performance testing), OWASP ZAP (security scanning), Lighthouse/axe (accessibility auditing), Jira/GitHub Issues (defect tracking).

Test Data: Seeded student, recruiter, and admin accounts; sample internship listings; sample resumes (PDF); a sample aptitude question bank with a known answer key.

7. Test Schedule

Milestones:

- Test case design: 28-Sep-2026
- Environment setup: 30-Sep-2026
- Test execution start: 01-Sep-2026
- Test execution end: 08-Oct-2026
- UAT: 09-Oct-2026 to 12-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	HEEMADHAWALA R	Maintain test plan, coordinate execution, review NFR/security coverage and consolidate status
Test Engineer	JOSHWIN PAUL	Design/execute functional and system tests for interfaces, listings, applications and core features.
Developer	HARSHITH	Support test design, analytics/test-module verification, RTM and defect verification.
Product Owner	KANISHQ SURENDRAN	Support authentication, integrations, dashboards and end-to-end workflows; prepare test data.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from the dev team
Test environment downtime	Maintain backup environment on a cloud VM
Dependency on Python analytics microservice availability	Engage module owners early; maintain mock/stub analytics responses
Dependency on third-party email/SMS provider	Use a sandbox/mock provider for OTP and notification tests

11. Assumptions & Dependencies

- MongoDB Atlas / self-hosted MongoDB instance will be stable and available
- Test data (student/recruiter/admin accounts, sample listings, resumes) will be provided before execution
- Email/SMS OTP provider sandbox credentials will be available
- Python analytics/scoring microservice will be deployed alongside the API in the test environment

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for more than 4 hours
- Build is too unstable (blocks more than 30% of test cases)

Resume testing if:

- Blocking defects are resolved
- Environment is stabilized

13. Test Case Management & Traceability

The RTM ensures mapping of SRS requirements to test cases. Examples:

- IPP-F-001 (Role-based registration) → TC-Auth-01
- IPP-F-002 (JWT authentication) → TC-Auth-02
- IPP-F-010 (Apply to internship) → TC-App-01
- IPP-F-011 (Update application status) → TC-App-02
- IPP-F-014 (Timed aptitude test) → TC-Test-01
- IPP-F-017 (Test analytics recording) → TC-Test-04
- IPP-F-019 (Placement analytics dashboard) → TC-Dash-02
- IPP-NF-001 (Response time) → TC-Perf-01
- IPP-SR-006 (RBAC enforcement) → TC-Sec-07

Full requirement-to-test-case mapping, including status (N/P/A), is maintained in the RTM (SRS Section 8) and updated throughout execution.

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed
- Defect density
- Defect aging
- Requirement coverage (against RTM)

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Date
QA Lead	HEEMADHAWALA R	16/9/2026
Dev Lead	HARSHITH	16/9/2026
Product Owner	KANISHQ SURENDRAN	16/9/2026